Decent Coaching Classes

IX CBSE

Mind Your Math VIII

Time : 3 Hours **Maximum Marks : 80**

General Instructions

1. All questions are compulsory.
2. The question paper has 5 sections: A, B, C, D and E.
3. Section A: 20 MCQs of 1 mark each. Section B: 5 very short answer questions of 2 marks each. Section C: 6 short answer questions of 3 marks each. Section D: 4 long answer questions of 5 marks each. Section E: 3 case-based questions of 4 marks each.
4. Graph paper is required for Question 32 of Section D.
5. Show all steps in your working. Use π = 22/7 wherever required.

SECTION A — Multiple Choice Questions 20 × 1 = 20

Choose the correct option.

1. The point (−3, 4) lies in the

   (a) I quadrant (b) II quadrant (c) III quadrant (d) IV quadrant

2. The distance of the point (3, −4) from the x-axis is

   (a) 4 units (b) 3 units (c) 5 units (d) −4 units

3. Which of the following points lies on the y-axis?

   (a) (4, 0) (b) (2, 2) (c) (0, −3) (d) (−1, 5)

4. The zero of the linear polynomial p(x) = 2x − 6 is

   (a) −3 (b) 6 (c) 12 (d) 3

5. Which of the following is a linear polynomial in x?

   (a) 3x − 5 (b) $x^2 + 1$ (c) 1/x + 2 (d) √x

6. If x = 2 is a zero of the polynomial 3x − k, then k =

   (a) 2 (b) 3 (c) 6 (d) −6

7. Which of the following is an irrational number?

   (a) √16 (b) √7 (c) 0.25 (d) 22/7

8. $2^3 \times 2^4$ is equal to

   (a) 2^{12} (b) 4^7 (c) 2^7 (d) 2

9. The value of $81^{1/4}$ is

   (a) 9 (b) 3 (c) 27 (d) 1/3

10. On rationalising the denominator, 1/(√3 − 1) becomes

    (a) (√3 − 1)/2 (b) √3 + 1 (c) (√3 + 1)/4 (d) (√3 + 1)/2

11. (x + 3)(x − 3) =

    (a) $x^2 - 9$ (b) $x^2 + 9$ (c) $x^2 - 6x + 9$ (d) $x^2 + 6x + 9$

12. If x + y = 7 and xy = 12, then $x^2 + y^2$ =

    (a) 37 (b) 19 (c) 25 (d) 49

13. Using a suitable identity, 101 × 99 =

    (a) 10001 (b) 9999 (c) 9899 (d) 10099

14. The angle subtended by a diameter of a circle at any point on the circle is

(a) 90° (b) 60° (c) 120° (d) 180°

15. The longest chord of a circle is its

(a) radius (b) arc (c) diameter (d) secant

16. An arc of a circle subtends an angle of 100° at the centre. The angle it subtends at a point on the remaining part of the circle is

(a) 100° (b) 200° (c) 25° (d) 50°

17. The circumference of a circle of radius 7 cm is (take $\pi = 22/7$)

(a) 44 cm (b) 154 cm (c) 49 cm (d) 22 cm

18. The perimeter of a rectangle of length 12 cm and breadth 8 cm is

(a) 96 cm (b) 20 cm (c) 80 cm (d) 40 cm

19. A die is rolled once. The probability of getting a prime number is

(a) 1/6 (b) 1/2 (c) 1/3 (d) 2/3

20. The 5th term of the AP 2, 5, 8, 11, … is

(a) 11 (b) 17 (c) 14 (d) 20

SECTION B — Very Short Answer Questions 5 × 2 = 10

21. Write the coordinates of the point which lies 3 units to the right of the y-axis and 5 units below the x-axis. **[2]**

22. If $p(x) = 2x - 3$, find $p(-1)$ and the zero of $p(x)$. **[2]**

23. Simplify: $27^{2/3} \times 5^0$. **[2]**

24. A bag contains 3 red and 5 blue balls. One ball is drawn at random. Find the probability that it is red. **[2]**

25. Find the 10th term of the AP 7, 10, 13, … **[2]**

SECTION C — Short Answer Questions 6 × 3 = 18

26. Express 0.777… in the form p/q, where p and q are integers and $q \neq 0$. **[3]**

27. Using a suitable identity, evaluate 102^2. **[3]**

28. A chord of length 16 cm is drawn in a circle of radius 10 cm. Find the distance of the chord from the centre. **[3]**

29. A rectangular field is 40 m long and 25 m wide. A path 2 m wide runs all around it on the outside. Find the area of the path. **[3]**

30. A die is thrown once. Find the probability of getting (i) an even number, (ii) a number greater than 4, (iii) a number less than 7. **[3]**

31. Which term of the AP 5, 11, 17, 23, … is 101? **[3]**

SECTION D — Long Answer Questions 4 × 5 = 20

32. Draw the graph of the linear equation $y = 2x - 1$.

(a) Complete the table of values for $x = -1, 0, 1, 2, 3$. [2]

(b) Plot the points on graph paper and join them to get the line. [2]

(c) From your graph, read the value of x when $y = 1$. [1]

33. If $x + 1/x = 4$, find the value of

(i) $x^2 + 1/x^2$ [3]

(ii) $x^4 + 1/x^4$ [2]

34. O is the centre of a circle and AB is a chord with ∠AOB = 110°. A point C lies on the major arc and a point D lies on the minor arc AB. Find (i) ∠ACB, (ii) ∠ADB, (iii) ∠OAB. **[5]**

35. A circular garden of radius 14 m is surrounded by a path 7 m wide. Find the area of the path and the cost of paving it at ₹ 50 per m^2. (Use π = 22/7) **[5]**

SECTION E — Case-Based / Integrated Questions 3 × 4 = 12

36. Case Study 1 — Taxi fare

A taxi service charges a fixed amount of ₹ 50 plus ₹ 12 for every kilometre travelled. Let x be the distance travelled in km.

(i) Write the fare as a linear polynomial in x. **[1]**

(ii) Find the fare for a journey of 8 km. **[1]**

(iii) Ravi paid ₹ 230. How many kilometres did he travel? **[2]**

37. Case Study 2 — Favourite sport

In a survey of 50 students, 20 chose cricket, 15 chose football, 10 chose hockey and 5 chose kabaddi as their favourite sport. One student is selected at random.

(i) Find the probability that the student chose cricket. **[1]**

(ii) Find the probability that the student did not choose football. **[1]**

(iii) Find the probability that the student chose hockey or kabaddi. **[2]**

38. Case Study 3 — Monthly savings

Riya saves ₹ 200 in the first month and increases her saving by ₹ 50 every month.

(i) Write the first four terms of the sequence of her monthly savings. **[1]**

(ii) How much does she save in the 12th month? **[1]**

(iii) Find her total savings in the first 12 months. **[2]**

— End of Question Paper —